

# WindAnalyzer

## First-Time User Guide

Guided Step-by-Step Documentation

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### Abstract

This document serves as a comprehensive, step-by-step guide for first-time users of the WindAnalyzer application. Available at <https://paulamacaira.shinyapps.io/WindAnalyzer/>, this tool provides powerful geographic and statistical analysis for wind data. This guide will walk you through the interface, data selection, and the various analytical components including clustering and time-series projections.

## Contents

|          |                                                            |          |
|----------|------------------------------------------------------------|----------|
| <b>1</b> | <b>Getting Started: Data Requirements</b>                  | <b>2</b> |
| 1.1      | Data File Structure . . . . .                              | 2        |
| 1.2      | Historical Wind Power Data (Mandatory) . . . . .           | 2        |
| 1.3      | Historical Wind Speed Data (Optional) . . . . .            | 2        |
| 1.4      | Projection Speed Data (Optional but Recommended) . . . . . | 2        |
| <b>2</b> | <b>Geographic Data Selection</b>                           | <b>2</b> |
| 2.1      | Generating Wind Speed Data Automatically . . . . .         | 3        |
| 2.2      | Understanding Bias Correction . . . . .                    | 3        |
| <b>3</b> | <b>Applying Methodology and Visualizing Data</b>           | <b>3</b> |
| 3.1      | Data Segregation Methods . . . . .                         | 3        |
| 3.2      | The “Apply” Action . . . . .                               | 4        |
| 3.3      | Dynamic Filters and Chart Visualization . . . . .          | 4        |
| <b>4</b> | <b>Future Projections and Exporting Data</b>               | <b>4</b> |
| 4.1      | Choosing the Number of Scenarios . . . . .                 | 4        |
| 4.2      | Downloading the Results . . . . .                          | 4        |

# 1 Getting Started: Data Requirements

To begin using WindAnalyzer, you need to provide data for the analysis. The tool operates using up to three datasets, though only the historical Wind Power data is strictly mandatory. If you lack historical wind speed data, the application features an automatic data generation capability based on geographic coordinates (detailed in **Geographic Data Selection** section).

If you wish to test the tool's capabilities before formatting your own data, WindAnalyzer provides three built-in demo spreadsheets. You can download these demo files from our GitHub repository: <https://github.com/gabriellsar/WindAnalyzer>.

When uploading your own data, the application only accepts files in .csv format. Ensure they follow the structure outlined below.

## 1.1 Data File Structure

All three required files (Wind Speed, Wind Power, and Projection Power) must follow the exact same columnar structure.

The files must contain the following three columns, named exactly as shown:

- **Date:** The date of the measurement in YYYY-MM-DD format.
- **Hour:** The hour of the measurement (integers from 0 to 23).
- **Speed** or **Power:** The numerical value of the wind speed or wind power generation for that specific date and hour.

## 1.2 Historical Wind Power Data (Mandatory)

This file contains the historical wind power generation data for your selected locations. The third column must be named **Power**.

## 1.3 Historical Wind Speed Data (Optional)

This file contains the historical wind speed measurements. The third column must be named **Speed**. If you do not have this file, the system can estimate historical wind speeds based on the locations you select on the map in the next step.

## 1.4 Projection Speed Data (Optional but Recommended)

This file contains the forecasted wind speed data for future periods. It is required to utilize the predictive analysis and simulate future power generation scenarios. The third column must be named **Speed**, and the dates should chronologically follow your historical data.

# 2 Geographic Data Selection

The first interactive module in WindAnalyzer is the “Map Input.” This section allows you to define the geographic coordinates for your analysis.

**Important Note:** If you already possess the historical “Wind Speed” data file (described in the previous section) and have uploaded it, you can skip this map selection step entirely. The system will use the data provided in your uploaded file.

However, if you do not have historical wind speed data, this step is mandatory, as the application will generate it for you based on the locations you select.

## 2.1 Generating Wind Speed Data Automatically

If you choose to generate the wind speed data automatically, you will use the interactive map and the accompanying form to define the parameters of the data extraction:

- **Latitude and Longitude:** You can manually type the geographic coordinates of your target location. Please note the valid geographic boundaries for this application: Latitude ranges from 7.65 (North) to -34.25 (South), and Longitude ranges from -75.95 (West) to -33.40 (East).
- **Period (Start and End Date):** Define the timeframe for the data generation.

**CRITICAL REQUIREMENT:** The date range you define here **must match exactly** the date range of the historical “Wind Power” file you intend to upload. If the periods do not align perfectly, the subsequent analyses (such as clustering) will fail or produce inaccurate results.

- **Rotor Height:** Input the height (in meters) of the wind turbine hub/rotor. Because wind speed varies significantly with altitude, providing an accurate rotor height is crucial for realistic power generation estimates.

## 2.2 Understanding Bias Correction

When WindAnalyzer automatically generates wind speed data, it fetches information from global meteorological datasets (such as MERRA-2). While robust, these datasets represent macro-level estimates and may have a slight deviation (bias) from the actual ground truth at your specific micro-location.

To improve the accuracy of your analysis, the form includes a Bias Correction configuration:

- **Apply Bias Correction?:** You can choose to enable or disable this feature. Current literature recommends using bias correction if the INMET station is at most 40 km away from the wind farm/turbine.
- **Average Correction Type:** The correction factor, derived from INMET station measurements, can be applied as: **Single Period** (the same applied to the entire series), **Monthly** (one for each month), **Hourly** (one for each hour) and **Monthly-Hourly** (one for each hour of each month). The calculation of the factors is done by the ratio between the average data from the INMET station and the average data from MERRA-2.

Once your data is successfully generated (and corrected) or your own files are uploaded, you are ready to proceed to the “Analysis” module.

## 3 Applying Methodology and Visualizing Data

### 3.1 Data Segregation Methods

To tailor the modeling to the specific characteristics of your location, you must select a Modeling Method. This choice dictates how the data is segregated before the statistical steps are applied:

- **Single Period:** The modeling is done globally, without any data segregation.
- **Monthly:** Data is segregated by month before applying the methods.
- **Hourly:** Data is segregated by hour of the day to carry out the modeling.
- **Monthly-Hourly:** The most detailed approach, segregating data by both month and specific time (hour).

## 3.2 The “Apply” Action

**Important:** Choosing a segregation method does not automatically update the analysis. You must always click the **Apply Methodology** button to execute the statistical modeling and generate the charts.

## 3.3 Dynamic Filters and Chart Visualization

Once the methodology is applied, WindAnalyzer generates multiple visualizations (like Density Curves and Time Series Projections).

The application will dynamically display interactive filters based strictly on the segregation method you chose:

- If you modeled by **Monthly**, a dropdown filter for the Month will appear above the charts.
- If you modeled by **Hourly**, a dropdown filter for the Hour will appear.
- If you modeled by **Monthly-Hourly**, both Month and Hour filters will become available.

Use these filters to navigate through the generated graphs, you can isolate specific segments of the data. For example, if you chose the “Monthly-Hourly” method, you can use the filters to view only the wind behavior at 14:00 (Hour filter) during August (Month filter).

## 4 Future Projections and Exporting Data

Once the statistical modeling is complete and validated, the final step in the WindAnalyzer workflow is to project the wind power scenarios and export your findings.

### 4.1 Choosing the Number of Scenarios

Before generating these projections, you must define the Number of Scenarios. This parameter dictates how many distinct, statistically possible future pathways the application will simulate.

- A higher number of scenarios provides a more robust and statistically significant spread of possible outcomes, though it may take slightly longer to compute.
- Once you select the desired number, ensure you execute the simulation to view the results.

### 4.2 Downloading the Results

WindAnalyzer allows you to easily extract the results of your modeling and simulations for use in external reports or further analysis. In the projection module, you will find options to download the following datasets in spreadsheet format:

1. **Mean wind generation time series:** A straightforward dataset containing the calculated average (mean) expected power generation across all simulated scenarios over the projected time period.
2. **Wind generation scenarios:** A comprehensive file containing the detailed data for every single scenario you chose to simulate.
3. **Wind speed time series:** The projected wind speed data used as the basis for the power simulations.

By downloading these files, you complete the WindAnalyzer workflow, moving from raw historical data to robust, actionable future projections.